

PROJECT FALCON

Wednesday Presentation Guide

Clear slide-by-slide script · Target duration: 9–11 minutes

Main message: Project FALCON is an affordable, local-first coastal monitoring buoy concept with a working software foundation. The funding request is intended to build, calibrate, and field-test the physical system. Do not claim that its wave prediction or sensors are already field validated.

Before You Present

- Assign one main speaker and one member who will operate the slides and dashboard.
- Speak slowly. Pause after every major idea instead of reading the slide word for word.
- Use the slide as visual evidence; use this guide to explain what the audience should understand.
- If interrupted by a question, answer briefly and return using the transition line.

Slide-by-Slide Script

Slide 1 — Project FALCON (35–45 seconds)

Goal: Introduce the team, the project, and the purpose of the presentation.

“Good day, everyone. We are the Project FALCON Research Group from Fullbright College. Project FALCON is our proposed smart buoy for observing coastal conditions in one specific location. In simple terms, it is a floating device with sensors, a small computer, solar power, and a dashboard. It will collect information near the deployment site and show the data in an easier form. Today, we will explain the problem, our proposed solution, what already works in software, and what we still need to build and test.”

Transition: “To understand why we designed FALCON, let us first look at the monitoring gap.”

Slide 2 — Why This Matters (50–60 seconds)

Goal: Explain the problem without criticizing PAGASA or claiming that no coastal information exists.

“Why is this project needed? We already have official forecasts, and we respect them. FALCON will not replace PAGASA or any official warning system. Our concern is that a forecast for a large area may not describe the exact condition at a small near-shore site. Professional monitoring equipment can also be expensive and difficult for a school or small local organization to maintain. So our question is simple: can we build a more affordable buoy that provides useful information for one

selected pilot site in Palawan?”

Transition: “Our response to this gap is a small monitoring system that can measure, process, store, and explain data locally.”

Slide 3 — Proposed Solution (55–65 seconds)

Point to Measure, Process Locally, Display and Alert, and Research Prediction in that order.

“Our proposed solution is a smart coastal buoy with four basic jobs. First, the sensors measure movement, water pressure, wind, location, temperature, and available power. Second, the ESP32 gathers the sensor readings, while the Orange Pi saves and processes them. The Orange Pi is simply the small onboard computer of the buoy. Third, the dashboard shows the information through numbers, graphs, history, and alerts. Fourth, after collecting enough real data, we will study whether the system can estimate wave height several minutes ahead. For now, the prediction is only a simulated research demonstration and not yet a real operational forecast.”

Transition: “The next image shows where these functions are located on the proposed buoy.”

Slide 4 — Upper Assembly and Sensor Levels (55–65 seconds)

Point from the top sensor deck downward. Do not discuss an underwater pointed section because it is not part of the approved visual.

“This image shows the parts above the water. At the top are the wind sensor, GPS, and antennas because they need open space to work properly. The solar panels are mounted on the tower and provide energy for the system. The large box in the middle is the upper electronics pod. It's a dry and protected box where we plan to place the ESP32, Orange Pi, storage, and other electronic boards. The motion sensor is attached to a rigid part of the structure so it measures the movement of the buoy itself. The round body at the bottom is the main float that keeps the structure above water. Its final balance and waterline will still be tested using the physical prototype.”

Transition: “Inside that structure, the sensors and power system follow this proposed electronics plan.”

Slide 5 — Hardware Concept (60–70 seconds)

Explain function, not every wire. Point first to sensors, then ESP32, Orange Pi, and finally the protected power chain.

“This diagram explains the job of each electronic component. The BNO085 measures the tilt and movement of the buoy. The Bar02 measures water pressure, which may help us estimate wave movement after proper calibration. GPS tells us the location, while the wind sensor measures wind speed and direction. The INA260 boards check the battery and solar power, and the MCP9808

checks the temperature inside the electronics box. All sensor readings go first to the ESP32. The ESP32 then sends the data to the Orange Pi for storage, processing, and display on the dashboard. The lower section shows how solar energy charges the battery and safely supplies power to the electronics. This is still a design guide; we will confirm the exact wiring using the real boards and their official datasheets.”

Transition: “To make this system practical, the electronics must also be organized for safe maintenance.”

Slide 6 — Maintainable Enclosure (50–60 seconds)

Point to the edge-computing zone, sensor-control zone, terminal area, then cable glands.

“This slide shows how we plan to arrange the electronics inside the upper box. The parts will be mounted on a removable tray, so we can inspect or replace them without removing the whole buoy structure. The Orange Pi side handles computing, while the ESP32 side handles sensor reading. Terminal blocks keep the wires organized instead of leaving many loose connections. Each cable will have a label on both ends, and cables entering the box will pass through waterproof locking glands. We will also separate signal wires from power wires to reduce electrical interference. Before actual deployment, we still need to test heat, waterproofing, wire clearance, and ease of maintenance.”

Transition: “With the hardware organized, this is how information travels through the system.”

Slide 7 — System Workflow (45–55 seconds)

Trace the arrows from sensors to ESP32, Orange Pi, database/API, then dashboard.

“This slide shows the flow of information. First, the sensors take measurements. Next, the ESP32 collects and checks those readings. It sends them to the Orange Pi as telemetry, which simply means organized sensor data. The Orange Pi saves the data, performs calculations, checks possible alerts, and sends the results to the dashboard. The dashboard then presents the information as readable values, graphs, and status messages. We call this local-first because the main processing and storage happen inside the buoy system, even without a continuous internet connection.”

Transition: “We have already implemented the software foundation for this workflow.”

Slide 8 — Current Progress (50–60 seconds)

Emphasize both the completed work and the honest limitation shown in the red/simulation notice.

“What have we finished so far? We already created the responsive dashboard, the basic ESP32 firmware, the data format used by the system, the simulator, the Orange Pi or laptop edge service, the local database, alerts, wiring plans, bill of materials, and project documentation. We also tested whether the firmware, backend service, and dashboard can build and operate in software. However,

we want to be honest: the current dashboard data are simulated. They do not yet come from a complete buoy in the sea. Our next stage is to connect the real sensors, calibrate them, build the buoy, and perform supervised field testing.”

Transition: “This clear separation between completed work and future validation is also part of our innovation approach.”

Slide 9 — Innovation (45–55 seconds)

Do not say that the team invented the buoy, AI, or sensors.

“We are not claiming that we invented the buoy, the sensors, or artificial intelligence. These technologies already exist. Our proposed innovation is how we combine them into one affordable and maintainable system for a specific local use. FALCON joins sensors, local processing, solar-power monitoring, data storage, a dashboard, and a future prediction model. It also clearly labels whether information is measured, simulated, estimated, or predicted. Our contribution will be proven only if the complete system is calibrated and tested for one selected coastal site.”

Transition: “The value of that integration depends on who will use it and what specific need it will serve.”

Slide 10 — Beneficiaries (40–50 seconds)

Name possible beneficiaries, then narrow the study to one primary partner.

“Several groups may benefit from this project. A coastal LGU or DRRM office may use the observation history and equipment alerts. Fisherfolk or small-boat users may benefit from easier access to information about one nearby site. Schools and researchers may use the platform and collected dataset for future studies. However, we will not claim that one prototype can immediately serve everyone. We will first choose one partner, one pilot location in Palawan, and one specific information need. If the first pilot is useful, the project can be improved and expanded later.”

Transition: “To reach that verified pilot, we propose a staged development plan.”

Slide 11 — Development Plan (45–55 seconds)

Move through the roadmap from design and procurement to controlled testing and reporting.

“Our estimated development time is eight to twelve months. First, we will finalize the design, exact components, safety requirements, and purchasing plan. Second, we will assemble the electronics and fabricate the buoy enclosure and structure. Third, we will calibrate the sensors and test power, waterproofing, and communication in controlled conditions. After those tests pass, we will conduct a supervised near-shore pilot. We will use the collected data to compare our AI model with a baseline, meaning a simple prediction method used as a reference. Finally, we will report the results, errors,

limitations, and recommended improvements.”

Transition: “Completing these stages requires support for more than electronics alone.”

Slide 12 — Preliminary Budget (45–55 seconds)

Point to the categories instead of reading every amount. Call the amount a planning range.

“Our preliminary budget is around seventy-two thousand to one hundred twenty-seven thousand pesos. Why is it a range? Because the exact marine-rated parts, fabrication cost, and supplier prices still need final quotations. The budget includes not only sensors and computers, but also solar power, battery, waterproof enclosure, buoy structure, mooring, safety materials, calibration, field testing, transportation, spare parts, and contingency. In other words, the request covers the complete process of building and testing a safe prototype. Before purchasing, we will submit an updated bill of materials with actual supplier prices.”

Transition: “With this support, these are the concrete results we intend to deliver.”

Slide 13 — Expected Result and Request (50–60 seconds)

End with the deliverables, specific request, and evidence-focused closing statement.

“If the project receives support, our goal is to produce one complete and documented physical prototype. We will deliver calibrated sensor results, tested solar and battery operation, a working local dashboard, stored coastal data, and supervised pilot-test findings. We will also compare the future AI model with a simple baseline to determine whether AI actually improves the prediction. We are requesting support for the components, fabrication, calibration, field testing, and technical guidance. To be clear, we are not asking you to trust an untested forecast. We are asking for the opportunity to build the system properly and collect evidence that will show whether it works. Thank you, and we are ready for your questions.”

Likely Questions and Simple Answers

Question	Suggested answer
Is the AI prediction already working?	“The interface and forecast pipeline work with simulated data, but the model is not yet field validated. We need calibrated local measurements before training and comparing it with a persistence baseline.”
What makes FALCON different?	“Its contribution is the affordable, local-first, and transparently validated integration of sensing, edge computing, power monitoring, prediction research, and displacement alerts for one defined Palawan use case.”

Will it replace PAGASA?	"No. FALCON is a research and localized observation tool. Official advisories and warnings remain the authority."
Why use both ESP32 and Orange Pi?	"The ESP32 handles reliable sensor reading. The Orange Pi handles storage, processing, the dashboard, and future model inference. Separating these roles improves reliability and maintainability."
How will you prove the wave measurements are accurate?	"We will calibrate the sensors, document the buoy and mooring response, and compare synchronized results with a suitable reference such as a pressure logger, staff gauge, video method, or validated wave instrument."
How will it operate without internet?	"Measurements are processed and stored locally. Internet or long-range communication can be used for remote access, but the core acquisition and archive do not depend on continuous cloud service."
Why is the budget a range?	"The exact marine-rated components, fabrication method, and suppliers still require final quotation. The range prevents us from presenting an invented exact amount before procurement."
What happens after funding?	"We finalize the design, purchase verified components, assemble and calibrate the system, perform controlled tests, conduct a supervised pilot, evaluate the model, and submit documented results."

Final Reminder

If you forget a line, return to this sentence: **"FALCON is a working software concept that needs funded physical development, calibration, and supervised field validation."**